

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

IR DINESH KUMAR(192312258) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Dinesh Kumar R

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

12. Design a machine learning model for spam email classification using labeled data. (5 Marks)

- Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
- Describe the preprocessing steps such as tokenization, stop-word removal, and

vectorization.

(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept *EnjoySport*.

(a) Explain the initialization of *S* (specific boundary) and *G* (general boundary).

(b) Show how *S* and *G* change with each training example.

(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).

(b) Explain feature scaling and handling missing data.

(c) Describe how overfitting can be detected and avoided.

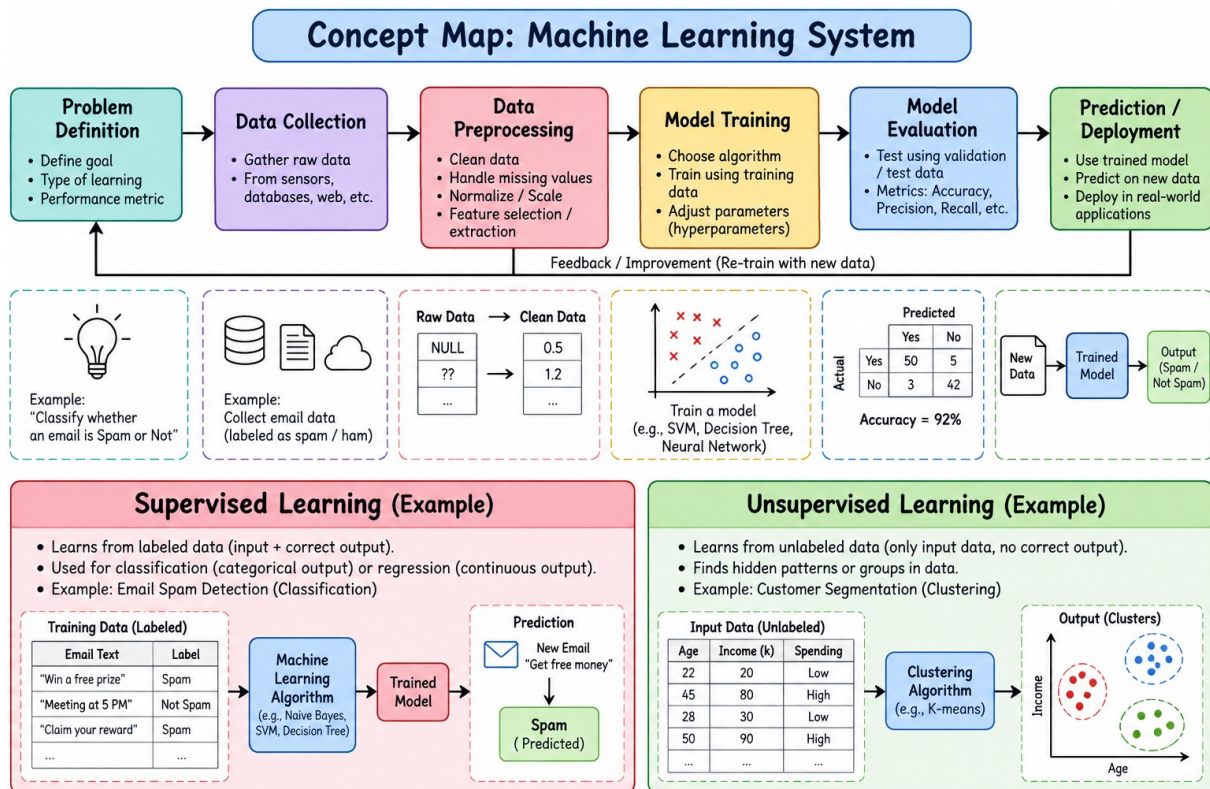
15. Explain the concept of clustering in unsupervised learning. (5 Marks)

(a) Describe the working of the K-Means algorithm with a step-by-step example.

(b) Explain how the value of *K* is chosen using the Elbow Method.

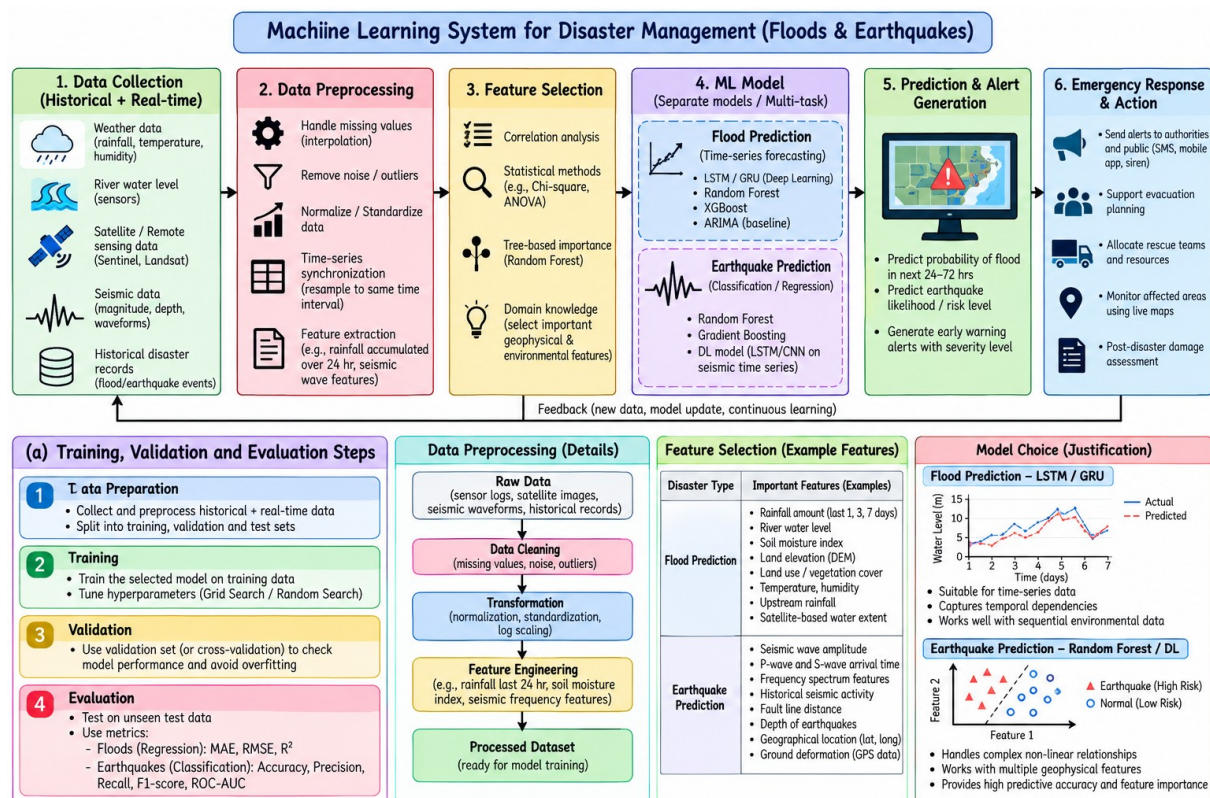
(c) Compare K-Means with Hierarchical Clustering.

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

- Describe the steps involved in training, validating, and evaluating the model.
- Explain how the system can assist in early warning and emergency response.

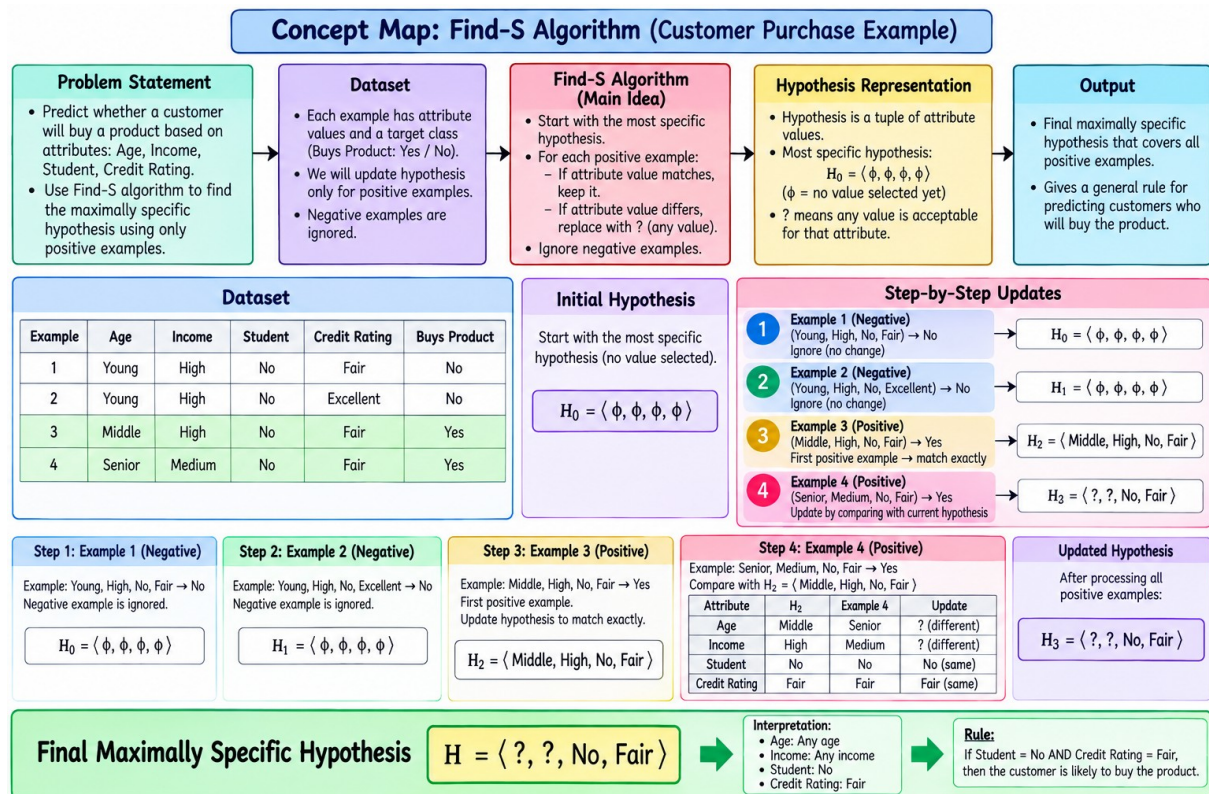


5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.

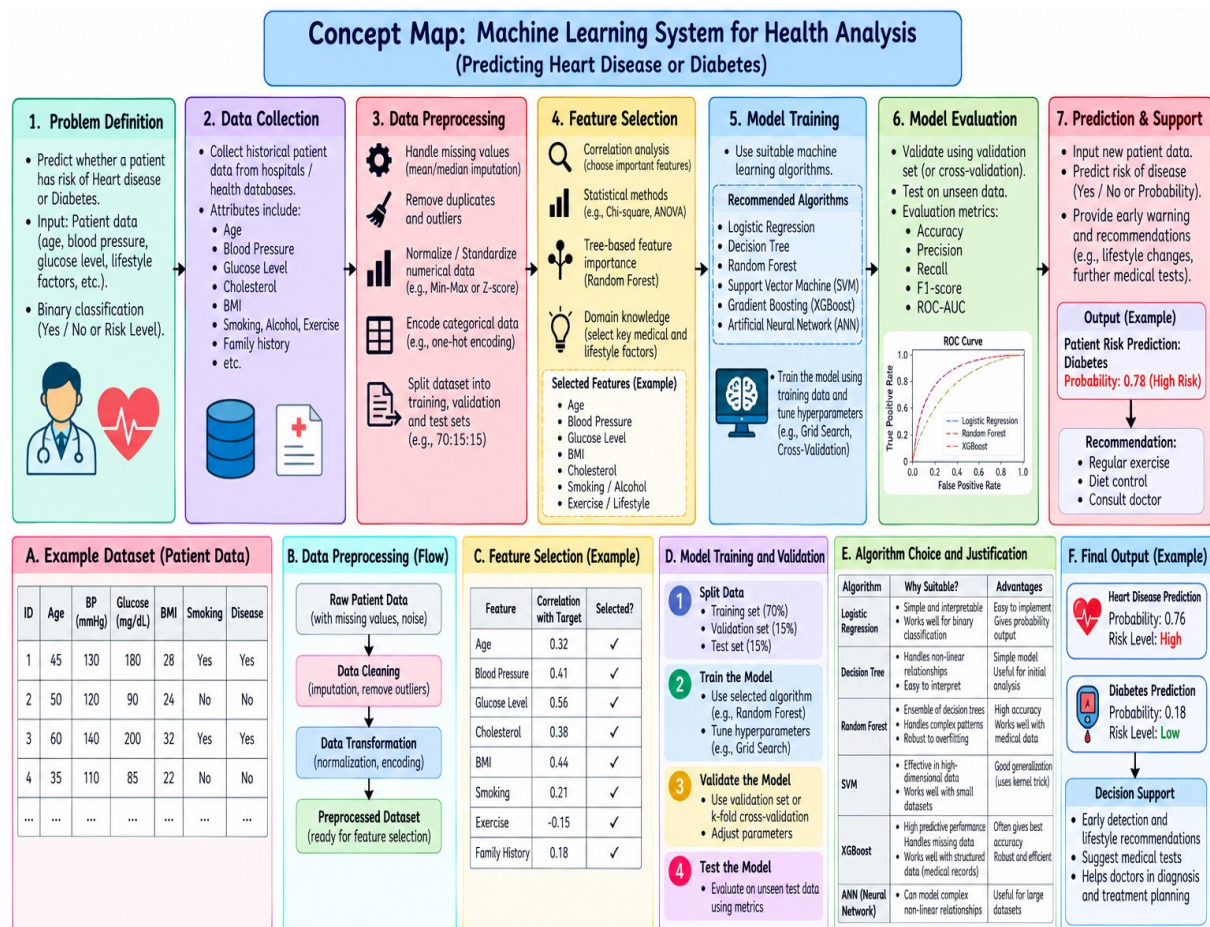
(5 Marks)

Dataset:

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10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions



A. Example Dataset (Patient Data)

ID	Age	BP (mmHg)	Glucose (mg/dL)	BMI	Smoking	Disease
1	45	130	180	28	Yes	Yes
2	50	120	90	24	No	No
3	60	140	200	32	Yes	Yes
4	35	110	85	22	No	No
...	...	...	...	...	...	...

B. Data Preprocessing (Flow)

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graph TD
    A[Raw Patient Data  
(with missing values, noise)] --> B[Data Cleaning  
(imputation, remove outliers)]
    B --> C[Data Transformation  
(normalization, encoding)]
    C --> D[Preprocessed Dataset  
(ready for feature selection)]
    
```

C. Feature Selection (Example)

Feature	Correlation with Target	Selected?
Age	0.32	✓
Blood Pressure	0.41	✓
Glucose Level	0.56	✓
Cholesterol	0.38	✓
BMI	0.44	✓
Smoking	0.21	✓
Exercise	-0.15	✓
Family History	0.18	✓

D. Model Training and Validation

- Split Data**
 - Training set (70%)
 - Validation set (15%)
 - Test set (15%)
- Train the Model**
 - Use selected algorithm (e.g., Random Forest)
 - Tune hyperparameters (e.g., Grid Search)
- Validate the Model**
 - Use validation set or k-fold cross-validation
 - Adjust parameters
- Test the Model**
 - Evaluate on unseen test data using metrics

E. Algorithm Choice and Justification

Algorithm	Why Suitable?	Advantages
Logistic Regression	Simple and interpretable Works well for binary classification	Easy to implement Gives probability output
Decision Tree	Handles non-linear relationships Easy to interpret	Simple model Useful for initial analysis
Random Forest	Ensemble of decision trees Handles complex patterns Robust to overfitting	High accuracy Works well with medical data
SVM	Effective in high-dimensional data Works well with small datasets	Good generalization (uses kernel trick)
XGBoost	High predictive performance Handles missing data Works well with structured data (medical records)	Often gives best accuracy Robust and efficient
ANN (Neural Network)	Can model complex non-linear relationships	Useful for large datasets

F. Final Output (Example)

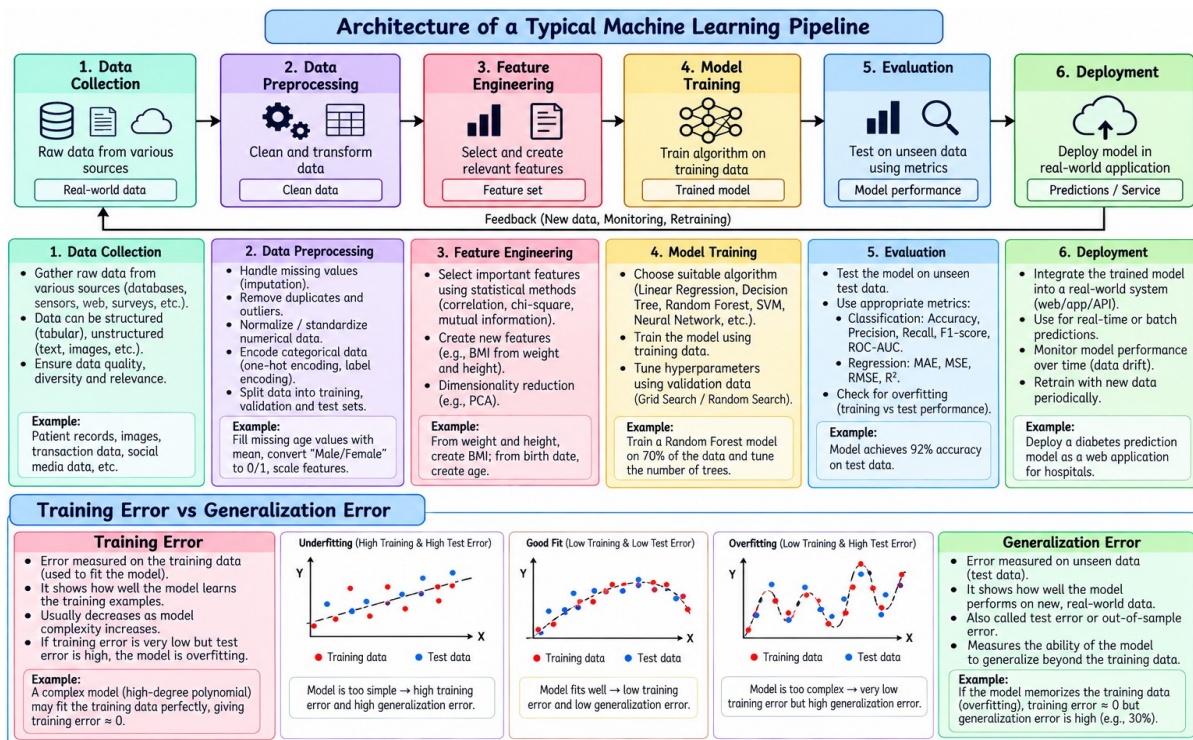
Heart Disease Prediction
Probability: 0.76
Risk Level: **High**

Diabetes Prediction
Probability: 0.18
Risk Level: **Low**

Decision Support

- Early detection and lifestyle recommendations
- Suggest medical tests
- Helps doctors in diagnosis and treatment planning

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand